

Week 4 - part 2

Quantum Computing

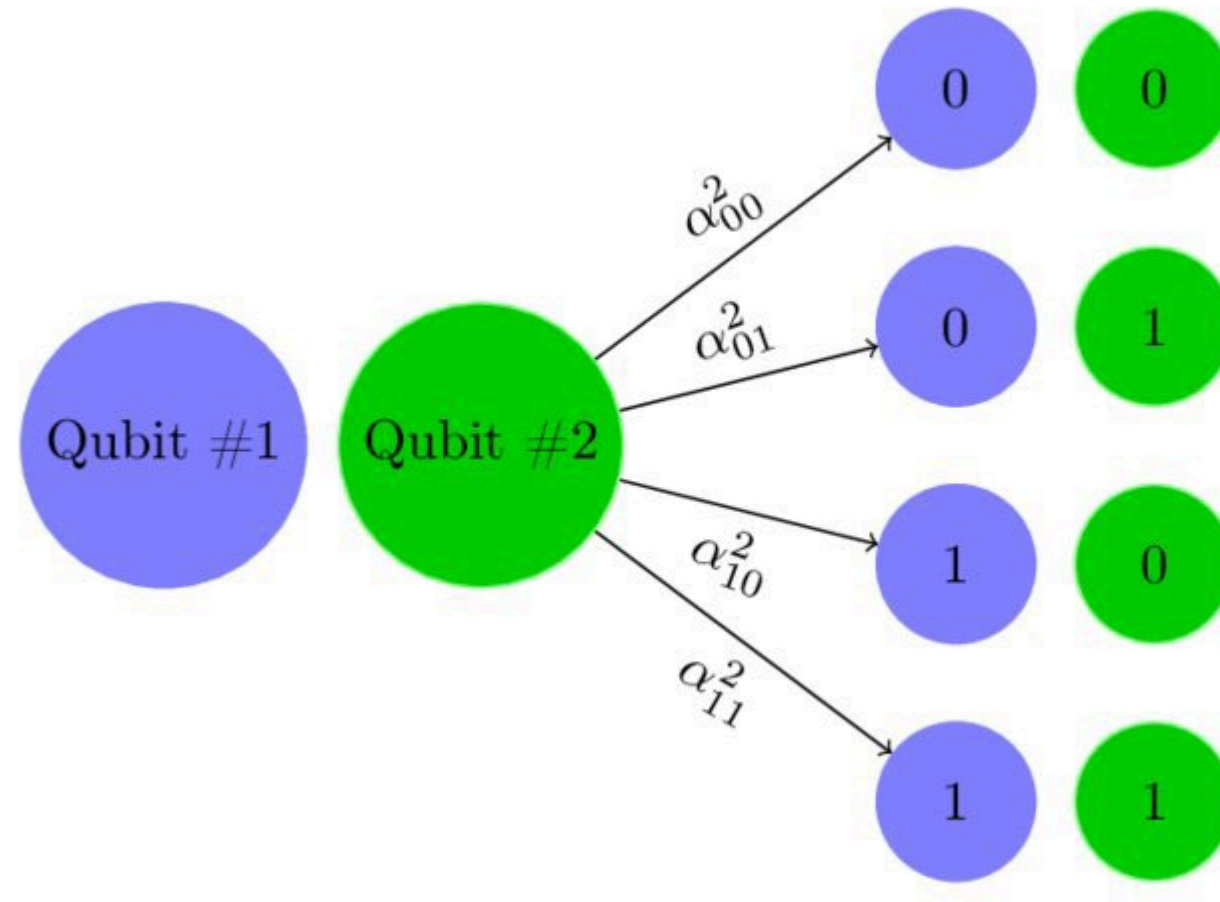
Quantum Logic and Gates

Logic for Multiple Qubits

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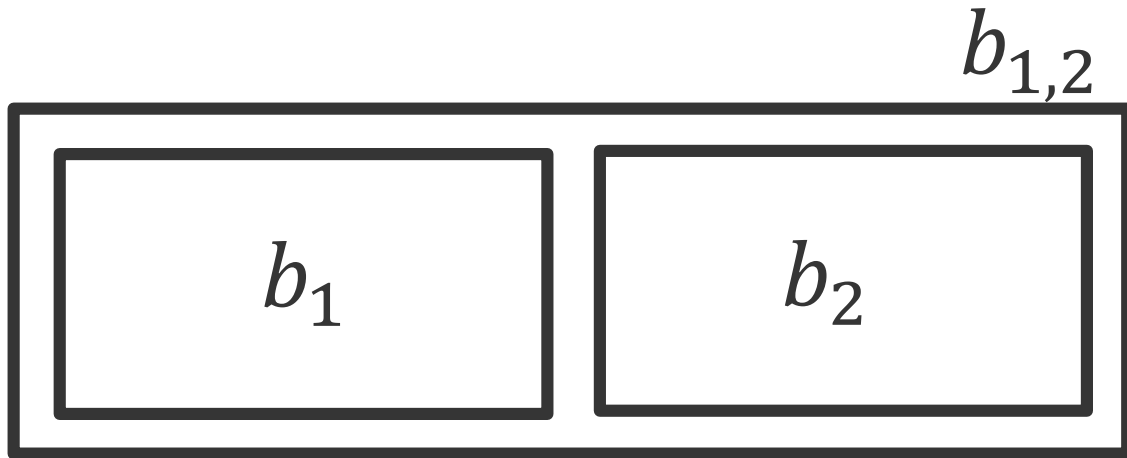
Wright State University

2 QUBIT LOGIC



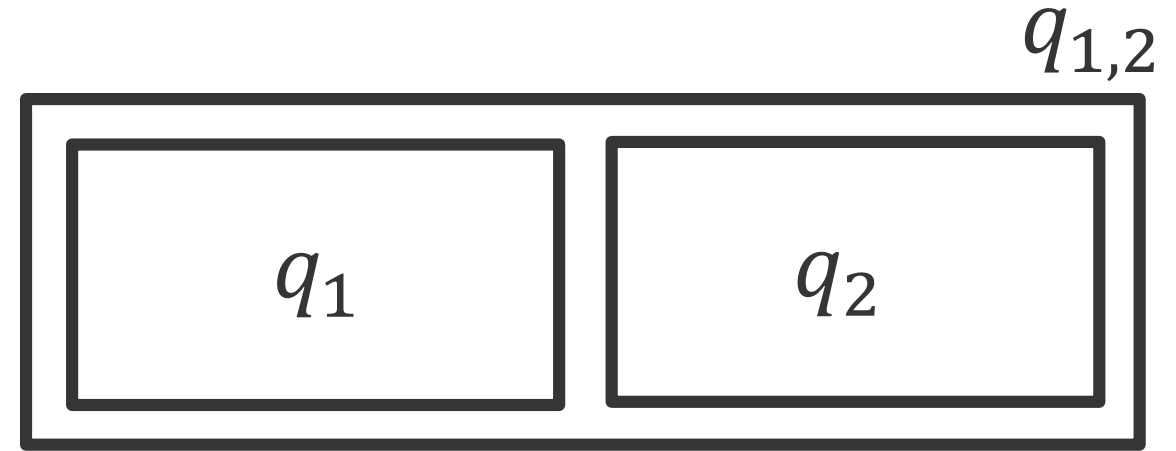
Multiple Qubits

Storing more information: multiple qubits



$b_{1,2} = 00 \text{ or } 01 \text{ or } 10 \text{ or } 11$

$$\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \text{ or } \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \begin{matrix} \text{--- "00"} \\ \text{--- "01"} \\ \text{--- "10"} \\ \text{--- "11"} \end{matrix}$$



$$q_{1,2} = \begin{pmatrix} c_{00} \\ c_{01} \\ c_{10} \\ c_{11} \end{pmatrix} \begin{matrix} \text{--- "00"} \\ \text{--- "01"} \\ \text{--- "10"} \\ \text{--- "11"} \end{matrix}$$

$$|c_{00}|^2 + |c_{01}|^2 + |c_{10}|^2 + |c_{11}|^2 = 1.$$

Measure "00" with prob $|c_{00}|^2$, etc.

Reading out more information: multiple qubits

$$q_{1,2} = \begin{pmatrix} c_{00} \\ c_{01} \\ c_{10} \\ c_{11} \end{pmatrix} \begin{matrix} \text{-- "00"} \\ \text{-- "01"} \\ \text{-- "10"} \\ \text{-- "11"} \end{matrix} \quad q_{1,2} = c_{00} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + c_{01} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} + c_{10} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} + c_{11} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

$$|c_{00}|^2 + |c_{01}|^2 + |c_{10}|^2 + |c_{11}|^2 = 1.$$

Measure "00" with prob $|c_{00}|^2$, etc.

Measure first qubit to be "0" with prob $|c_{00}|^2 + |c_{01}|^2$, with post-measurement state

$$q_1 = \frac{1}{\sqrt{|c_{00}|^2 + |c_{01}|^2}} \left(c_{00} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + c_{01} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \right)$$

Quick review: Dirac notation

"Ket" notation: $|\cdot\rangle$ denotes a column vector

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, |q\rangle = \begin{pmatrix} c_0 \\ c_1 \end{pmatrix} = c_0|0\rangle + c_1|1\rangle$$

Let $|\psi\rangle$ denote a column vector, where $c_i \in \mathbb{C}^{2^n}$:

$$|\psi\rangle = \begin{pmatrix} c_0 \\ c_1 \\ \vdots \\ c_{2^n-2} \\ c_{2^n-1} \end{pmatrix} = \sum_{i=0}^{2^n-1} c_i |i\rangle$$

We will sometimes denote the i th basis state in integer form for compactness, rather than in binary

n is the number of qubits in the quantum state.

Quick review: Dirac notation

$$|\psi\rangle = \begin{pmatrix} c_{00} \\ c_{01} \\ c_{10} \\ c_{11} \end{pmatrix} \begin{matrix} \text{--- "00"} \\ \text{--- "01"} \\ \text{--- "10"} \\ \text{--- "11"} \end{matrix} = c_{00} |00\rangle + c_{01} |01\rangle + c_{10} |10\rangle + c_{11} |11\rangle \\ = c_0 |0\rangle + c_1 |1\rangle + c_2 |2\rangle + c_3 |3\rangle$$

Ket Notation:

Shorthand notation for column vectors.

Used a lot in the literature.

Sometimes more compact.

$$|\varphi\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} |000\rangle + \frac{1}{\sqrt{2}} |110\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |6\rangle)$$

Quick review: Dirac notation

Inner product:

Input: two vectors

Output: complex number

Recall * denotes
complex conjugate

$$((a_0 \dots a_{2^n-1}), (c_0 \dots c_{2^n-1})) = \sum_i a_i^* c_i = (a_0^* \dots a_{2^n-1}^*) \begin{pmatrix} c_0 \\ \vdots \\ c_{2^n-1} \end{pmatrix}$$

"Bra": $\langle\psi|$ denotes a row vector

It is the complex conjugate transpose of $|\psi\rangle$

$$\langle\psi| = (c_0^* \ c_1^* \ \dots \ c_{2^n-1}^*)$$

$\langle\varphi|\psi\rangle$, also denoted $\langle\varphi, \psi\rangle$, is thus the inner product of $|\varphi\rangle$ and $|\psi\rangle$:

$$\begin{aligned} \langle\varphi|\psi\rangle &= \sum_i a_i^* c_i \\ \langle\psi|\psi\rangle &= \sum_i c_i^* c_i = \sum_i |c_i|^2 = 1 \end{aligned}$$

Quick review: orthogonality

- Two vectors are orthogonal if their inner product is 0: $\langle \varphi | \psi \rangle = 0$.
- Example: Verify the computational basis states are orthogonal:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

- The norm of a vector is given by

$$\| |q\rangle \| = \sqrt{\langle q | q \rangle} = \left(\sum_i c_i^* c_i \right)^{1/2}$$

- A vector is a unit vector or normalized if $\| |q\rangle \| = 1$.
- Normalize a vector by dividing by its norm: $\frac{|q\rangle}{\| |q\rangle \|}$

Remember What is a Tensor Product?

If we have a [basis](#) for the vector spaces, and the vector space is finite-dimensional, we can represent the vectors in terms of components under those basis vectors:

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}, \mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_m \end{bmatrix}.$$

where each notation stands for the sum $\mathbf{v} = v_1 \mathbf{e}_1 + v_2 \mathbf{e}_2 + \cdots + v_n \mathbf{e}_n$.

A tensor is then a map $T(\mathbf{v}, \mathbf{w})$ that works as above, returning a scalar and is linear in both of its arguments. Such a tensor can be represented using a matrix multiplication:

$$T(\mathbf{v}, \mathbf{w}) = \mathbf{v}^T \mathbf{T} \mathbf{w}$$

where the superscripted T denotes the [matrix transpose](#) which sends the vector $\mathbf{v}$ to its [dual vector](#).

Given two vectors, we can form a tensor of their own from them rather naturally using the **outer product**, which is denoted $\mathbf{v} \otimes \mathbf{w}$ and equals $\mathbf{v} \mathbf{w}^T$. This tensor comes out as the matrix

$$\mathbf{v} \otimes \mathbf{w} = \begin{bmatrix} v_1 w_1 & v_1 w_2 & \cdots & v_1 w_m \\ v_2 w_1 & v_2 w_2 & \cdots & v_2 w_m \\ \vdots & \vdots & \ddots & \vdots \\ v_n w_1 & v_n w_2 & \cdots & v_n w_m \end{bmatrix}$$

Tensor products

We write $|p\rangle \otimes |q\rangle$, $|p\rangle|q\rangle$, $|pq\rangle$, or $|p, q\rangle$ to represent the tensor product.

Input: states $|p\rangle, |q\rangle$ with dimensions m, n respectively

Output: $|p\rangle \otimes |q\rangle$ with dimension mn

$$|p\rangle \otimes |q\rangle = \begin{pmatrix} p_0 q_0 \\ p_0 q_1 \\ \vdots \\ p_0 q_{n-1} \\ \vdots \\ p_{m-1} q_{n-1} \end{pmatrix}$$

Now you can see
why 2 qubits cover
a state space of
dimension 4, 3
qubits cover
dimension 8, and n
qubits cover
dimension 2^n

Tensor products

Example:

$$|0\rangle \otimes |0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = |00\rangle = |0\rangle$$

$$|1\rangle \otimes |0\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = |10\rangle = |2\rangle$$

A (tensor) product state:

$$\begin{aligned} & (a_0|0\rangle + a_1|1\rangle)(c_0|0\rangle + c_1|1\rangle) \\ &= a_0c_0|00\rangle + a_0c_1|01\rangle + a_1c_0|10\rangle + a_1c_1|11\rangle \end{aligned}$$

Example: Bell states

Can the following state be represented as a product state?

$$\frac{1}{\sqrt{2}} |00\rangle + \frac{1}{\sqrt{2}} |11\rangle$$

No!

This is an entangled state. It cannot be separated into a product of two states.

This state can exhibit “nonlocal” correlations.

Entanglement

Unentangled

Alice

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Bob

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)(|0\rangle + |1\rangle)$$

$$q_{AB} = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \begin{matrix} \text{-- "00"} \\ \text{-- "01"} \\ \text{-- "10"} \\ \text{-- "11"} \end{matrix}$$

$$\Pr(00) = \Pr(01) = \Pr(10) = \Pr(11) = \frac{1}{4}.$$

Entangled

Alice

Bob

cannot write separately

$$\frac{1}{\sqrt{2}} |00\rangle + \frac{1}{\sqrt{2}} |11\rangle$$

$$q_{AB} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} \begin{matrix} \text{-- "00"} \\ \text{-- "01"} \\ \text{-- "10"} \\ \text{-- "11"} \end{matrix}$$

$$\Pr(00) = \Pr(11) = \frac{1}{2}.$$

$$\Pr(01) = \Pr(10) = 0.$$